Faculty of Engineering Department of

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Graduation Project 1

Academic Advising and Student Progress Monitoring System

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Introduction

Academic advising is an important part of higher education. It helps students make informed academic decisions, set educational goals, and overcome challenges that may affect their progress. However, managing advising records, support plans, and follow-up activities can be difficult when information is stored in separate documents or systems.

The Academic Advising and Student Progress Monitoring System is a web-based application designed to organize advising activities, manage student cases, and monitor progress toward academic goals.

Scientific Problem :

Managing academic advising activities can be challenging when student concerns, advising records, and support plans are stored separately. This can make it difficult for academic advisors to review students' histories, track their progress, and follow up on previously established goals.

Therefore, there is a need for an integrated system that organizes advising information and supports continuous monitoring of student progress.

Proposed Solution :

The proposed solution is a web-based system that provides an integrated environment for managing academic advising activities and monitoring student progress. It enables students to record their academic concerns, request advising sessions, and follow their goals, while academic advisors can manage student cases, document meetings, create support plans, and review progress.

The system will maintain organized records of advising activities to facilitate follow-up and help advisors understand each student's situation over time.

Project Objectives :

The main objective of this project is to develop a web-based system that organizes academic advising activities and supports student progress monitoring.

The specific objectives are to:

1. **Organize student advising records in one centralized system.**
2. **Enable students to record academic concerns and request advising sessions.**
3. **Help advisors document meetings and create student support plans.**
4. **Track student progress toward academic goals.**
5. **Facilitate follow-up and improve access to advising information.**

Litrature Review :

Several platforms have been developed to support academic advising and student success. The following section reviews three examples, highlighting their main features and limitations in relation to the proposed project.

Similar Systems :

1. EAB Navigate360 :

- Features: Student profiles, advising appointments, case management, alerts, and reports.
- Limitations: Its broad range of features may make implementation and customization complex for a small project.

2. Starfish :

- **Features: Appointment scheduling, student referrals, alerts, and communication between students and advisors.**
- **Limitations: Its available functions and integrations can depend on the institution's configuration and connected systems.**

3. Stellic :

- **Features: Degree planning, progress tracking, academic reports, and advising appointments.**
- **Limitations: Its main emphasis is academic planning and degree requirements rather than the full documentation of advising cases and support sessions.**

Feature	Navigate 360	Starfish	Stellic	Proposed System
Record student academic concerns	✓	✓	✓	✓
Document advising meetings	✓	✓	✓	✓
Create and manage support plans	✓	✓	✓	✓
Link support plans to individual academic goals	✓	✓	✓	✓
Combine all advising steps in one focused workflow	✓	✓	✓	✓
Designed specifically around this project's defined requirements	✗	✗	✗	✓

Project Implementation Phases :

The project will be implemented through the following phases:

Requirements Analysis: Identify the system requirements, users, and main functions.

System Design: Design the system architecture, database, and user interfaces.

Database and Backend Development: Create the SQL Server database and implement the application logic using ASP.NET Core.

Frontend Development: Develop the user interfaces and the functions required by students and academic advisors.

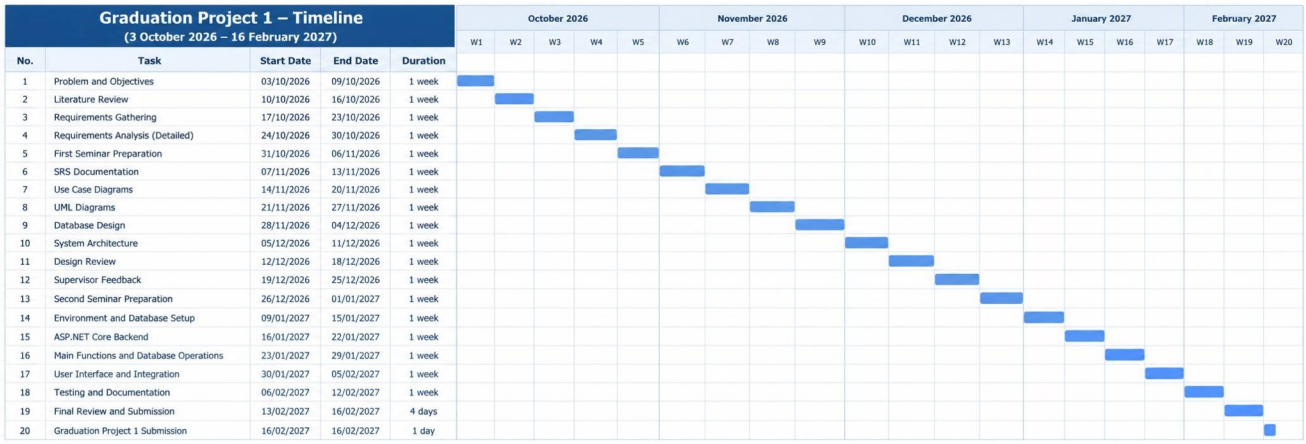
Integration and Testing: Integrate the system components, test the main functions, and fix identified errors.

Documentation and Evaluation: Document the implemented system and evaluate it against the defined requirements.

Project Implementation Phases :

- **ASP.NET Core**: To develop the web application and implement its main functionality.
- **C#:** To write the application logic and backend code.
- **SQL Server:** To store and manage student information, advising requests, meeting records, and progress data.
- **Entity Framework Core:** To facilitate communication between the application and the database.
- **HTML, CSS, and JavaScript:** To build interactive and user-friendly web interfaces.
- **Visual Studio:** To develop, debug, and test the application.
- **Git and GitHub:** To manage source code and track project changes.

Gantt Chart :



Graduation Project 1 – Timeline (3 October 2026 – 16 February 2027)				
No.	Task	Start Date	End Date	Duration
1	Problem and Objectives	03/10/2026	09/10/2026	1 week
2	Literature Review	10/10/2026	16/10/2026	1 week
3	Requirements Gathering	17/10/2026	23/10/2026	1 week
4	Requirements Analysis (Detailed)	24/10/2026	30/10/2026	1 week
5	First Seminar Preparation	31/10/2026	06/11/2026	1 week
6	SRS Documentation	07/11/2026	13/11/2026	1 week
7	Use Case Diagrams	14/11/2026	20/11/2026	1 week
8	UML Diagrams	21/11/2026	27/11/2026	1 week
9	Database Design	28/11/2026	04/12/2026	1 week
10	System Architecture	05/12/2026	11/12/2026	1 week
11	Design Review	12/12/2026	18/12/2026	1 week
12	Supervisor Feedback	19/12/2026	25/12/2026	1 week
13	Second Seminar Preparation	26/12/2026	01/01/2027	1 week
14	Environment and Database Setup	09/01/2027	15/01/2027	1 week
15	ASP.NET Core Backend	16/01/2027	22/01/2027	1 week
16	Main Functions and Database Operations	23/01/2027	29/01/2027	1 week
17	User Interface and Integration	30/01/2027	05/02/2027	1 week
18	Testing and Documentation	06/02/2027	12/02/2027	1 week
19	Final Review and Submission	13/02/2027	16/02/2027	4 days
20	Graduation Project 1 Submission	16/02/2027	16/02/2027	1 day